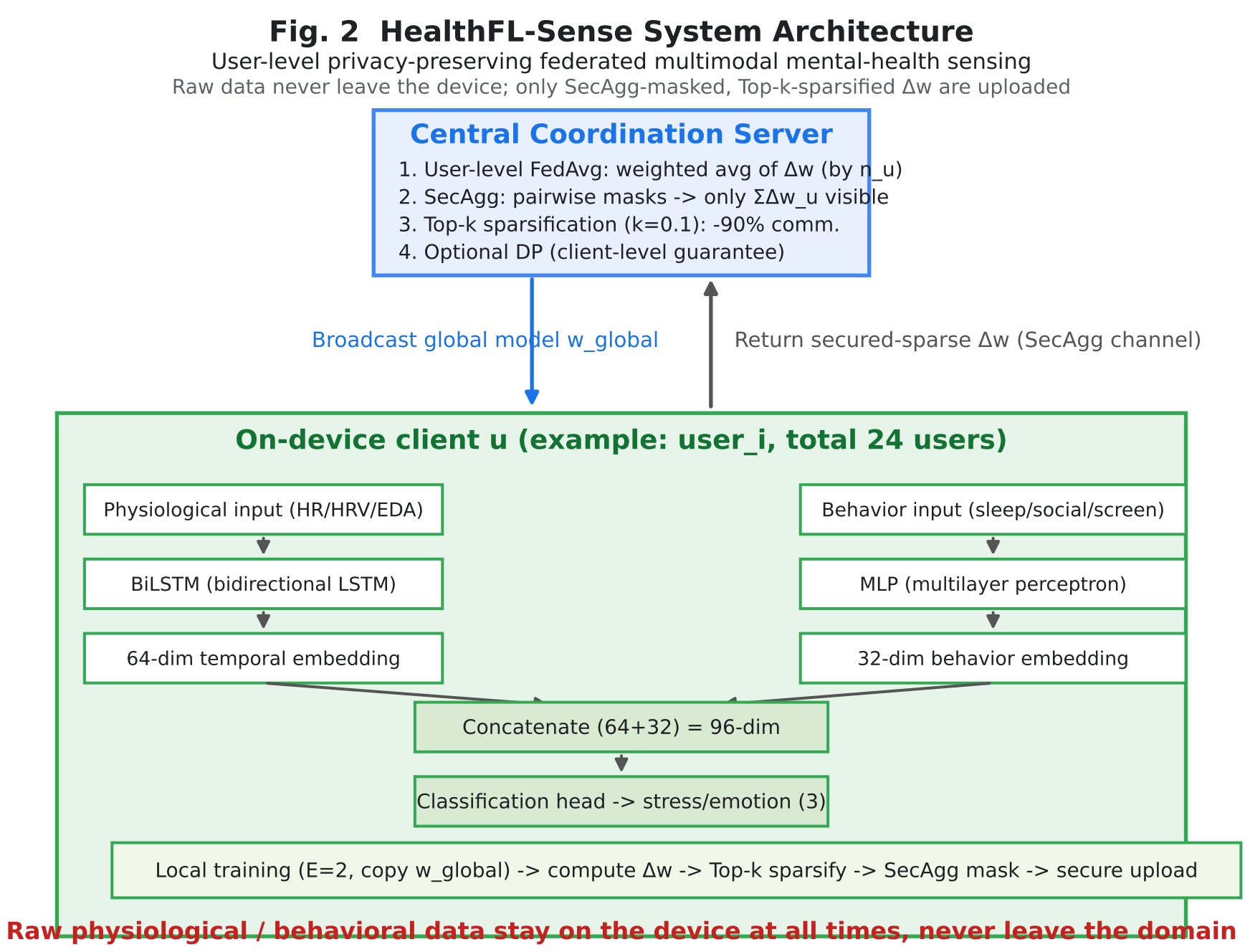
Fig. 2 HealthFL-Sense System Architecture
User-level privacy-preserving federated multimodal mental-health sensing
Raw data never leave the device; only SecAgg-masked, Top-k-sparsified Δw are uploaded
Central Coordination Server
1. User-level FedAvg: weighted avg of Δw (by n_u)
2. SecAgg: pairwise masks -> only ΣΔw_u visible
3. Top-k sparsification (k=0.1): -90% comm.
4. Optional DP (client-level guarantee)
Broadcast global model w_global
Return secured-sparse Δw (SecAgg channel)
On-device client u (example: user_i, total 24 users)
Physiological input (HR/HRV/EDA)
BiLSTM (bidirectional LSTM)
64-dim temporal embedding
Behavior input (sleep/social/screen)
MLP (multilayer perceptron)
32-dim behavior embedding
Concatenate (64+32) = 96-dim
Classification head -> stress/emotion (3)
Local training (E=2, copy w_global) -> compute Δw -> Top-k sparsify -> SecAgg mask -> secure upload
Raw physiological / behavioral data stay on the device at all times, never leave the domain